

CRICKET SHOT CLASSIFIER

ML-Based Cricket Shot Type Recognition System

[Pull Shot](#) | [Leg Glance-Flick](#) | [Drive](#) | [Sweep](#)

[Kaggle Dataset](#) · [HOG Features](#) · [scikit-learn](#) · [Streamlit Cloud](#)

SUBMISSION DETAILS

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Course Title: Artificial Intelligence (ML; DL; Communication)

Batch: Batch II

Assignment URL: <https://student.tlms.navttc.gov.pk/assignment/69cfbf53658b656360a776d3>

Dataset

Kaggle (aneesh10)

Classes

4 Shot Types

GitHub

Najam000

Streamlit

Cloud Deployed

GitHub:
[https://github.com/Najam000/Cricket_shots_classifier-](https://github.com/Najam000/Cricket_shots_classifier)

Streamlit Cloud:
<https://cricket-shot-classifier.streamlit.app>

1. Project Overview

Cricket Shot Classifier is a supervised machine learning system that automatically identifies the type of batting shot played by a cricketer from a single image. Using HOG (Histogram of Oriented Gradients) feature extraction combined with multiple scikit-learn classifiers, the system classifies shots into four categories with high accuracy.

Problem Statement

Manual identification of cricket shots is time-consuming, requires deep expert knowledge, and cannot scale to modern broadcast volumes. An automated classifier enables coaches, broadcasters, and analysts to process player footage rapidly and at scale.

Real-World Applications

- Player coaching — automated feedback on batting technique and shot selection
- Broadcast analytics — real-time shot labelling during live cricket matches
- Fantasy cricket platforms — shot-level player profiling and statistics
- Talent scouting — data-driven evaluation of a batsman's shot variety
- Self-training apps — players test and improve their own shot technique

2. Dataset

The dataset is sourced from Kaggle and contains approximately 4,000 labelled images of cricket batsmen across four batting shot categories. Each class contains roughly 1,000 images captured from varied angles, lighting conditions, and player stances — covering both left-handed and right-handed batsmen.

Property	Value
Source	Kaggle — aneesh10 / cricket-shot-dataset
Dataset URL	https://www.kaggle.com/datasets/aneesh10/cricket-shot-dataset/data
Total Images	Approximately 4,000 images
Images per Class	Approximately 1,000 per shot type
Number of Classes	4 (Pull Shot, Leg Glance-Flick, Drive, Sweep)
Image Format	JPG / PNG
Task Type	Multi-class Image Classification
Train / Test Split	80 % training 20 % testing (stratified)

Shot Class Descriptions

Shot Type	Description
Pull Shot	Horizontal bat shot against short-pitched deliveries, pulling the ball to the leg side.
Leg Glance / Flick	Delicate wrist-guided deflection that angles the ball fine towards the leg side.
Shot Type	Description

Drive	Classical full-face stroke through the off or on side against full-pitched deliveries.
Sweep	Cross-bat shot played on one knee, sweeping spin deliveries to the leg side.

3. Methodology

Feature Extraction — HOG (Histogram of Oriented Gradients)

Each image is converted to a compact HOG descriptor rather than using raw pixel values. HOG captures edge orientations and gradient magnitudes — highly effective for human body pose and motion analysis. The resulting feature vector contains approximately 1,764 dimensions per image.

Pipeline: Input Image → RGB to Grayscale → Resize to 64 × 64 → HOG Extraction (9 orientations, 8 × 8 cells, 2 × 2 blocks) → 1,764-dim feature vector → StandardScaler normalisation

Notebook Stages (following Imports.docx template)

#	Stage	Description	Key Tools
1	Imports & Setup	Load all libraries, set seeds, configure display	numpy, sklearn, skimage, matplotlib
2	Load Dataset	Walk class folders, extract HOG feature per image	PIL, skimage.feature.hog
3	EDA	Class distribution, sample images, feature	matplotlib, seaborn, pandas
4	Preprocessing	80/20 stratified split + StandardScaler on features	StandardScaler, train_test_split
5	Baseline Models	Train 7 classifiers, compare train and test acc.	SVM, RF, GBM, ET, LR, KNN, NB
6	Cross-Validation	5-fold stratified CV on top 3 models	StratifiedKFold, cross_val_score
7	GridSearchCV	Hyperparameter tuning of the best CV model	GridSearchCV
8	Final Evaluation	Confusion matrix, classification report	sklearn.metrics
9	Save Model	Pickle model and scaler; reload to verify	pickle

Classifiers Compared in Baseline Stage

Model	Configuration	Notes
SVM (RBF Kernel)	C=1, gamma=scale	Excellent with HOG in high-dimensional space
Random Forest	100 estimators	Robust ensemble; handles non-linearity well
Gradient Boosting	100 estimators, lr=0.1	Strong generalisation via sequential boosting
Extra Trees	100 estimators	Faster randomised tree ensemble
Logistic Regression	max_iter = 1000	Linear baseline with good interpretability
K-Nearest Neighbours	k = 5	Instance-based learner; simple and effective
Model	Configuration	Notes

Gaussian Naive Bayes	Default parameters	Probabilistic baseline model
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4. Deployment

Streamlit Web Application

The trained model and scaler are packaged as a Streamlit web application. Users upload any cricket shot image and receive an instant prediction with confidence scores and a probability distribution chart for all four classes. The app is hosted on Streamlit Cloud and linked to the GitHub repository.

Item	Details
Streamlit URL	https://cricket-shot-classifier.streamlit.app
GitHub Project	https://github.com/users/Najam000/projects/1
App Script	app.py
Model File	cricket_shot_model.pkl
Scaler File	cricket_shot_scaler.pkl
Dependencies	streamlit, scikit-learn, scikit-image, numpy, Pillow, matplotlib

App Features

- Upload a JPG or PNG image — HOG features are extracted instantly
- Predicted shot type displayed with confidence percentage
- Probability bar chart showing scores for all four shot classes
- Shot description tab with cricketing context for each category
- Model info tab showing the full pipeline and stage descriptions
- Dataset reference card linking directly to the Kaggle source page

How to Run Locally

- Step 1.** Download the dataset from Kaggle; place it in cricket-shot-dataset/
- Step 2.** Open and run Cricket_Shot_Classifier.ipynb end-to-end in Jupyter Notebook
- Step 3.** This generates cricket_shot_model.pkl and cricket_shot_scaler.pkl
- Step 4.** Launch the app with: streamlit run app.py
- Step 5.** To deploy online: push all files to GitHub and connect to Streamlit Cloud

5. Results & Evaluation

The following metrics are generated after running the complete notebook pipeline. Exact numerical values depend on the dataset version, but the expected test accuracy exceeds 85 % with the tuned SVM (RBF) or Random Forest.

Metric	Value / Description
Feature Extraction	HOG — approximately 1,764 features per 64 × 64 image

Train / Test Split	80 % training / 20 % testing (stratified by class)
Cross-Validation	5-fold Stratified KFold on the top 3 baseline models
Tuning Method	GridSearchCV with inner StratifiedKFold cross-validation
Expected Test Acc.	Greater than 85 % after hyperparameter tuning
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score per class
Confusion Matrix	Normalised per-class recall + raw counts side-by-side

Visualisations Generated by the Notebook

File Name	Description
eda_class_distribution.png	Bar chart and pie chart showing class balance across four shot types
eda_sample_images.png	4 × 5 grid displaying five sample images per shot class
eda_hog_stats.png	HOG feature distribution histograms separated by class
baseline_comparison.png	Grouped bar chart of all seven model train and test accuracies
cv_boxplot.png	5-fold CV accuracy distribution boxplot for the top 3 models
gridsearch_results.png	GridSearchCV top-15 parameter combinations bar chart
confusion_matrix.png	Raw count matrix alongside normalised confusion matrix
per_class_accuracy.png	Per-class accuracy bar with overall accuracy reference line

6. Project Deliverables & Links

Deliverable	Link / File Name
Project Title	Cricket Shot Classifier — ML-Based Shot Type Recognition
Dataset (Kaggle)	https://www.kaggle.com/datasets/aneesh10/cricket-shot-dataset/data
Notebook File	Cricket_Shot_Classifier.ipynb
Python App File	app.py (Streamlit deployment script)
GitHub Project	https://github.com/users/Najam000/projects/1
Streamlit Cloud	https://cricket-shot-classifier.streamlit.app
Assignment URL	https://student.tlms.navttc.gov.pk/assignment/69cfbf53658b656360a776d3

Submitted by	Trainer	Course
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	Batch II	2024 – 25
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